

The all-order 1D chart theorem for smooth data

Claude, with Daniel Murfet

2026-09-07

Abstract

The all-order expansion of the one-dimensional chart integral (unit 96) is restated with hypotheses only on the regularity of the chart data: for $\xi, \eta \in C^R(\mathbb{R})$, $R \geq 1$,

$$\int_0^b u^h e^{-\beta(Nu^k)^2} \eta(u) e^{\beta(Nu^k)\xi(u)} du = \sum_{j < R} c_j N^{-(p+j/k)} + O(N^{-(p+R/k)}),$$

where the c_j are the jet coefficients of unit 96 evaluated at the Taylor data $x_i = \xi^{(i)}(0)/i!$, $y_i = \eta^{(i)}(0)/i!$. The order- R Taylor remainders on $[0, b]$ and the Lipschitz bound at 0 are extracted from Mathlib's `taylor_mean_remainder_bound` and compactness of $[0, b]$. Zero sorry/axiom.

1 The things formalised

```
noncomputable def taylorData (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (i :  $\mathbb{N}$ ) :  $\mathbb{R}$  :=
  iteratedDeriv i f 0 / (i.factorial :  $\mathbb{R}$ )
theorem taylorWithinEval_eq_taylorData (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (n :  $\mathbb{N}$ ) (b :  $\mathbb{R}$ ) (hb : 0 < b)
  (hf : ContDiff n f) (u :  $\mathbb{R}$ ) :
  taylorWithinEval f n (Icc 0 b) 0 u =  $\sum_{i < n+1}$  taylorData f i * u ^ i
theorem exists_taylor_remainder_bound (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (n :  $\mathbb{N}$ ) (b :  $\mathbb{R}$ ) (hb : 0 < b)
  (hf : ContDiff (n + 1) f) :
  K :  $\mathbb{R}$ , 0 < K  $\rightarrow \forall u \in \text{Icc } (0 : \mathbb{R}) \text{ b},$ 
  |f u -  $\sum_{i < n+1}$  taylorData f i * u ^ i| ≤ K * u ^ (n + 1)
theorem exists_lipschitz_bound (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (b :  $\mathbb{R}$ ) (hb : 0 < b) (hf : ContDiff 1 f) :
  L :  $\mathbb{R}$ , 0 < L  $\rightarrow \forall u \in \text{Icc } (0 : \mathbb{R}) \text{ b}, |f u - f 0| \leq L * u$ 
theorem oneDChart_all_orders_smooth (b p :  $\mathbb{R}$ ) (h k R :  $\mathbb{R}$ ) (h : 0 < ) (hb : 0 < b)
  (hk : 0 < k) (hp : ((h :  $\mathbb{R}$ ) + 1) / k = p) (hR : 0 < R)
  ( :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (h : ContDiff R ) (h : ContDiff R ) :
  K' :  $\mathbb{R}$ , N :  $\mathbb{N}$ , 1 ≤ N  $\rightarrow$ 
  |oneDScale b N h k
    -  $\sum_{j < R}$  jetCoeff p k (chartJet (taylorData ) (taylorData ) R) j
    * N ^ (-(p + (j :  $\mathbb{N}$ ) / k))|
  K' * N ^ (-(p + (R :  $\mathbb{N}$ ) / k))
```

2 Mathematics

Mathlib's Taylor theorem gives, for $f \in C^{n+1}$ on $[a, b]$ and $x \in [a, b]$, $|f(x) - T_n f(x)| \leq C(x - a)^{n+1}/n!$ whenever the $(n + 1)$ -st derivative within $[a, b]$ is bounded by C . With $a = 0$ and $f \in C^{n+1}(\mathbb{R})$ the within-derivatives agree with the global ones (`iteratedDerivWithin_eq_iteratedDeriv`, using `uniqueDiffOn_Icc`), the $(n + 1)$ -st derivative is continuous hence bounded on the compact interval, and the Taylor polynomial $T_n f(u) =$

$\sum_{i \leq n} \frac{f^{(i)}(0)}{i!} u^i$ is exactly the Taylor sum of unit 95. The Lipschitz bound is the $n = 0$ case. Feeding both into `oneDChart_all_orders` with the Taylor data gives the theorem; measurability of ξ, η comes from continuity.

3 Paper-facing meaning

Together with unit 96's closed form $c_j = k^{-1} \sum_{m < R} \frac{\beta^m}{m!} e_{j,m} S_{p+j/k+m}(\xi(0))$, this is a complete formal statement of the one-dimensional all-order fluctuation expansion under the paper's own hypotheses (smooth ξ, η near the exceptional divisor), with an explicit finite- R remainder. No claim is made about convergence of the full inverse- N series.

4 Lean notes

- `taylorWithinEval` lives in `Mathlib.Analysis.Calculus.Taylor`, which the project's import closure did not yet contain; import it explicitly.
- A line break inside a `by` block nested in a `rw [...]` list is a parse error; hoist the term into a named `have`.
- `ContDiff (n + 1) f` with `n :` elaborates the order as `((n + 1 : ) : WithTop Ⓜ)`, whereas `taylor_mean_remainder_bound` wants `(n : WithTop Ⓜ) + 1`; `exact_mod_cast` bridges both directions.
- `Finset.sum_range_one + iteratedDeriv_zero` reduce the $n = 0$ Taylor polynomial to $f(0)$ in one `simp` only.